Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030001

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Query 1: -CREATE DATABASE

```
CREATE DATABASE IF NOT EXISTS bookflow7_db;
USE bookflow7_db;
```

Output: -

1	16:06:18	CREATE DATABASE IF NOT EXISTS bookflow7_db	1 row(s) affected	0.015 sec
2	16:06:21	USE bookflow7_db	0 row(s) affected	0.000 sec

Query 2: - CREATE TABLE

```
2520030001-Aswitha
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output: -

3	16:09:19	CREATE TABLE bank_transactions (txn_id INT PR...	0 row(s) affected	0.063 sec

Query 3: - Insert and Display

```sql
INSERT INTO bank_transactions VALUES
(101,'Aswitha','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Adhritha','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Nihitha','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Ananya','Vizag','Deposit',8000,'2024-01-10'),
(105,'Rushika','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Priyadarshini','Vizag','Deposit',15000,'2024-01-12'),
(107,'Amulya','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Akshaya','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Srujana','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Varshini','Vijayawada','Deposit',11000,'2024-01-16');
SELECT * FROM bank_transactions;
```

Output: -

4	16:12:54	INSERT INTO bank_transactions VALUES (101,'Aswitha','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Adhrith...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
5	16:13:06	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Aswitha	Hyderabad	Deposit	5000.00	2024-01-05
102	Adhritha	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Nihitha	Vijayawada	Deposit	12000.00	2024-01-08
104	Ananya	Vizag	Deposit	8000.00	2024-01-10
105	Rushika	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Priyadarshini	Vizag	Deposit	15000.00	2024-01-12
107	Amulya	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Akshaya	Hyderabad	Deposit	9000.00	2024-01-14
109	Srujana	Vizag	Withdrawal	4000.00	2024-01-15
110	Varshini	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Query 4: - SUM()

```sql
2520030001-Aswitha
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;
```

Output: -

6	16:15:55	SELECT SUM(amount) AS Total_Amount FROM ban...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

	Total_Amount
▶	70500.00

Query 5: - AVG()

```
2520030001-Aswitha
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;
```

Output: -

7 16:18:11 SELECT AVG(amount) AS Average_Transaction F... 1 row(s) returned 0.015 sec / 0.000 sec

Result Grid | Filter Rows:

	Average_Transaction
▶	7050.000000

Query 6: -MAX()

```
2520030001-Aswitha
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;
```

Output: -

8 16:19:46 SELECT MAX(amount) AS Highest_Transaction FR... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid | Filter Rows:

	Highest_Transaction
▶	15000.00

Query 7:- MIN()

```
2520030001-Aswitha
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```

Output: -

9	16:21:29	SELECT MIN(amount) AS Lowest_Transaction FR...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows:

Lowest_Transaction
1000.00

Query 8: -COUNT()

```
2520030001-Aswitha
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Output: -

10	16:23:22	SELECT COUNT(*) AS Total_Transactions FROM ...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows:

Total_Transactions
10

Query 9: -WHERE + SUM()

```
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output: -

11	16:25:08	SELECT SUM(amount) AS Total_Deposit FROM ba...	1 row(s) returned	0.000 sec / 0.000 sec

Total_Deposit
60000.00

Query 10: - GROUP BY

```
2520030001-Aswitha
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

12 16:26:59 SELECT branch_name, SUM(amount) AS Total... 3 row(s) returned 0.000 sec / 0.000 sec

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

Query 11: - GROUP BY + HAVING

```
2520030001-Aswitha
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;
```

Output: -

13 16:28:47 SELECT branch_name, SUM(amount) AS Total... 2 row(s) returned 0.000 sec / 0.000 sec

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

Query 12: - GROUP BY + ORDER BY

```
2520030001-Aswitha
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```

Output: -

14	16:30:48	SELECT branch_name,	SUM(amount) AS Total...	3 row(s) returned	0.016 sec / (

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

Query 13: - GROUP BY + HAVING + ORDER BY

```
2520030001-Aswitha
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```

Output: -

15	16:32:30	SELECT branch_name,	COUNT(*) AS Total_Tr...	3 row(s) returned	0.016 sec / 0.000 sec

Result Grid | Filter Rows:

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

Query 14: - WHERE + COUNT()

```
ORDER BY Total_Transactions DESC;
SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type='Withdrawal';
```

Output: -

16	16:34:22	SELECT COUNT(*) AS Withdrawals FROM bank_tr...	1 row(s) returned

Result Grid

	Withdrawals
▶	4

Query 15: - WHERE + AVG()

```
SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output: -

18	16:36:57	SELECT AVG(amount) AS Avg_Deposit FROM ban...	1 row(s) returned

Result Grid

	Avg_Deposit
▶	10000.000000

Query 16: - GROUP BY + MAX()

```
2520030001-Aswitha
SELECT branch_name,
       MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

19	16:39:15	SELECT branch_name, MAX(amount) AS High...	3 row(s) returned

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

Query 17: - GROUP BY + MIN()

```
2520030001-Aswitha
SELECT branch_name,
       MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

20 16:40:59 SELECT branch_name, MIN(amount) AS Lowe... 3 row(s) returned

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Query 18: - WHERE + ORDER BY

```
2520030001-Aswitha
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```

Output: -

21 16:42:50 SELECT * FROM bank_transactions WHERE amou... 4 row(s) returned

Result Grid | Filter Rows: | Edit: | Export/Import

txn_id	customer_name	branch_name	transaction_type	amount	transaction_
106	Priyadarshini	Vizag	Deposit	15000.00	2024-01-12
103	Nihitha	Vijayawada	Deposit	12000.00	2024-01-08
110	Varshini	Vijayawada	Deposit	11000.00	2024-01-16
108	Akshaya	Hyderabad	Deposit	9000.00	2024-01-14
NULL	NULL	NULL	NULL	NULL	NULL

Query 19: - GROUP BY + COUNT()

```
2520030001-Aswitha
SELECT branch_name,
       COUNT(*) AS Customer_Count
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

22 16:44:47 SELECT branch_name, COUNT(*) AS Custome... 3 row(s) returned

branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

Query 20: - HAVING + COUNT()

```
2520030001-Aswitha
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) > 3;
```

Output: -

23 16:51:28 SELECT branch_name, COUNT(*) AS Total_Tr... 1 row(s) returned

branch_name	Total_Transactions
Hyderabad	4

Query 21: - GROUP BY + AVG()

```
2520030001-Aswitha
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

24 16:53:14 SELECT branch_name, AVG(amount) AS Avg_... 3 row(s) returned 0.0(

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

Query 22: - WHERE + GROUP BY + HAVING

```
2520030001-Aswitha
SELECT branch_name,
       SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```

Output: -

25 16:54:58 SELECT branch_name, SUM(amount) AS Total... 2 row(s) returned 0.000 se

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

Query 23: - GROUP BY + HAVING + ORDER BY

```
2520030001-Aswitha
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

Output: -

26 16:56:27 SELECT branch_name, AVG(amount) AS Avg_... 2 row(s) returned

	branch_name	Avg_Amount
▶	Vizag	9000.000000
	Vijayawada	8000.000000